

PROFESSIONAL SUMMARY

Strategic Senior Laboratory Scientist with over 15 years' expertise in molecular diagnostics, virology, and GMP-compliant biomanufacturing. Proven leader in harmonizing global diagnostic standards (WHO), optimizing workflows (20 % faster batch releases) and delivering zero non-conformities in UKAS/GMP audits. Adept at aligning technical innovation with public-health priorities through cross-functional collaboration, high-containment pathogen management and data-driven process improvements.

Demonstrated capacity to mentor and develop multidisciplinary teams, fostering a culture of continuous improvement and rigorous quality standards. Published author of four WHO Technical Reports and frequent presenter at international virology conferences.

Core Competencies

Molecular & Virology

qPCR/RT-qPCR (Roche Cobas 6800)
ddPCR, ELISA (Dynex/Liaison XL)
Viral Propagation (CL3)
LIMS Automation

Quality & Compliance

ISO 13485/17025 & GMP/GLP
UKAS/OCABR Audit Leadership
CAPA, Risk Mitigation & IVDR
CE-Marked Reagent Release

Leadership & Strategy

WHO International Standardization
Multi-site Project Management
Cross-functional Team Leadership
Stakeholder Consensus Building

Data & Digital Tools

Advanced Excel, Prism, Minitab
GitHub, Jira, Azure Fundamentals
ReactJS Basics

Regulatory

IVDR Technical Documentation
Regulatory Submissions & Reporting
Quality System Documentation

Communication

Scientific Writing & SOPs
Training Delivery & Presentations
Stakeholder Reporting & Presentations

Selected Achievements

Global Diagnostic Harmonization (WHO)

Led and coordinated international studies involving 30+ laboratories across five continents to establish WHO primary NAT standards for WNV, HSV, HIV-2 and HAV. Developed standardized protocols and data-analysis frameworks that reduced inter-lab variability by 15%. Co-authored four WHO Technical Reports (2017–2020) and delivered findings at three major international virology symposia.

Operational Excellence (Anthony Nolan, 2023–2024)

Redesigned Dynex ELISA and LIMS workflows, boosting throughput by 25% and cutting manual data entry by 30%. Achieved zero major findings in the 2024 UKAS audit.

Pandemic Response & Resilient Delivery

- Coordinated production of a 7-member respiratory virus reference panel (3,000+ vials) during COVID-19, supporting 10+ vaccine programs despite 40% staff shortages.
- Maintained ISO 17025 plasma pool screening during Brexit demand surges through inventory optimization and proactive equipment maintenance.

Strategic Infrastructure Development

Directed Class 3 lab refurbishment for WNV standard production, integrating Paul Ehrlich Institute calibrants to meet WHO timelines.

Professional Experience

Senior Laboratory Scientist

Anthony Nolan Research Institute, London Aug 2023 – Feb 2024

- Optimized ABO blood-group testing on the Immunocor Echo platform, reducing error rates from 1.5% to 0.5% through comprehensive protocol standardization and staff retraining.
- Enhanced laboratory efficiency by streamlining LIMS data-logging workflows, resulting in a daily time saving of approximately 2 hours for technical teams.
- Contributed to the successful 2024 UKAS audit, achieving zero major findings by ensuring strict adherence to quality standards and continuous process improvement.

Scientist (HEO), Infectious Disease Diagnostics

NIBSC (MHRA), South Mimms Mar 2013 – Aug 2022

- Managed the end-to-end production of CE-marked reagents under ISO 13485 and GMP standards, maintaining 100% compliance across nine consecutive audits.
- Introduced droplet digital PCR (ddPCR) assays for vaccine stability testing, doubling sensitivity for low-level viral detection and enhancing quality control measures.
- Led WHO NAT standardization projects, including the upgrade of Containment Level 3 (CL3) facilities and fostering global stakeholder consensus through collaborative studies.
- Directed the refurbishment of SAPO4 Schedule 5-compliant CL3 laboratories for West Nile Virus standard production, integrating Paul Ehrlich Institute calibrants to meet WHO timelines.

Healthcare Scientist Practitioner

UK Health Security Agency, Colindale Jan 2010 – Jan 2011

- Performed RT-qPCR diagnostics for H1N1 pandemic response under CPA standards, ensuring rapid and accurate testing during a critical public health emergency.
- Authored emergency Standard Operating Procedures (SOPs) and trained 12 staff members, enhancing the laboratory's preparedness and response capabilities.

Earlier Roles:

- **Laboratory Manager**, *Smiths Detection*, **Biochemistry Technician**, *Bio Products Laboratory*, **Experimental Officer**, *Covance*. Full employment history available upon request.

Education

- **BSc (Hons) Medical Biochemistry**, *Brunel University London*, 2001 – 2005
- **Diploma in Professional Development**, *Northwick Park Hospital*, 2003 – 2004

Publications & Affiliations

Co-author, WHO Technical Reports: West Nile Virus RNA (2020), Herpes Simplex Virus DNA (2020), HIV-2 RNA (2018), Hepatitis A Virus RNA (2017)

Memberships

- Clinical Virology Network
- MHRA Diagnostics Working Group
- UKAS Best Practice Forum

Availability: Immediate

References: Available upon request